



TITLE PAGE

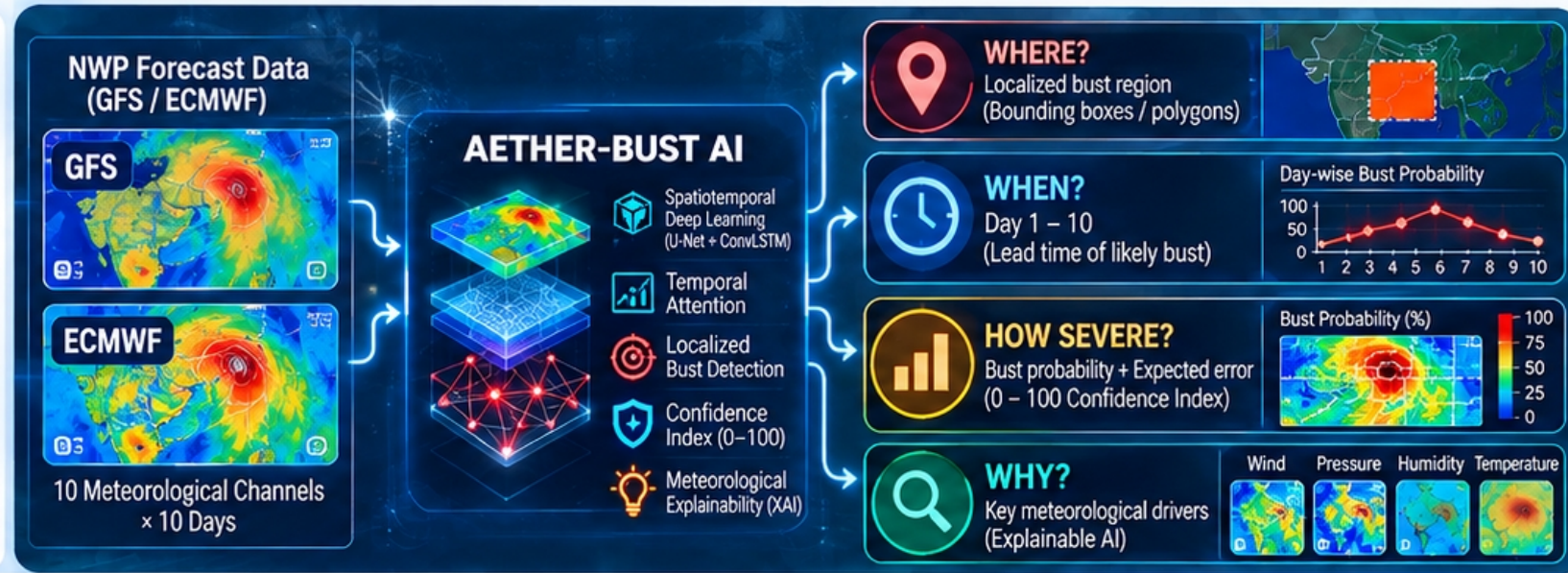
- **Problem Statement ID** : [SIH26079](#)
- **Problem Statement Title** : [AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts](#)
- **Theme** : [Smart Automation](#)
- **PS Category** : [Software](#)
- **Team ID** : [128249](#)
- **Team Name (Registered on portal)** : [TechBytes](#)





Proposed Solution

AETHER-BUST is an AI-based system that predicts **when** and **where** medium-range weather forecasts are likely to fail (forecast busts). It uses multi-parameter meteorological data from GFS/ECMWF and a spatiotemporal deep learning model to provide **localized bust probability, expected error** and **explainable reasons**.



01

Proposed Solution



- AI system to **predict when and where** medium-range weather forecasts are likely to fail (forecast busts).
- Uses **multi-parameter meteorological data** (Day 1–10) from GFS/ECMWF.
- Provides **localized bust probability**, expected error and confidence score with explainable reasons.

02

Detailed Explanation of the Proposed Solution



- Uses a **spatiotemporal deep learning model** (U-Net + ConvLSTM + temporal attention).
- Input: 10 meteorological channels × 10 forecast days (GFS/ECMWF).
- Outputs: **bust probability map**, expected error map, confidence index (0–100) and key meteorological drivers (XAI).
- Provides **localized bust masking** (bounding boxes / polygons).

03

How it addresses the problem



- Identifies potential **forecast failures** before they occur.
- Localizes **high-risk regions** for Day 1–10 forecasts.
- Helps **meteorological agencies focus verification and issue early alerts**.
- Supports better decision-making for disaster preparedness, aviation, agriculture, water management, etc.

04

Innovation & Uniqueness



- Spatiotemporal BustNet** with causal attention.
- Confidence index (0–100)** instead of only point prediction.
- Explainable AI** with 10 meteorological drivers (Grad-CAM + SHAP).
- Localized bust masking** and polygon-based high-risk areas.
- Operationally usable outputs** (maps, alerts, GIS exports).

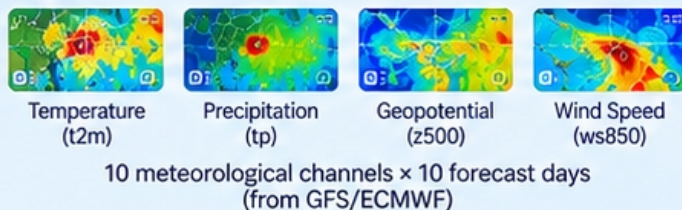


Detailed Explanation

AETHER-BUST uses a **spatiotemporal deep learning model** (U-Net + ConvLSTM with temporal attention) to predict forecast busts by learning patterns from multi-parameter weather data. It analyzes **10 meteorological channels** for the next 10 days and outputs **localized bust probability, expected error** and **explainable meteorological drivers**.



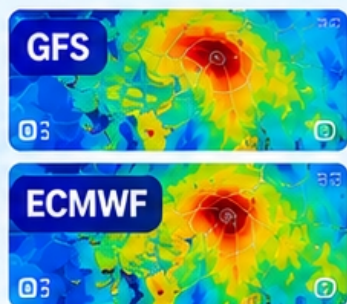
Input Data (Multi-parameter NWP)



System Architecture & Workflow

1

Data Input & Preprocessing

NWP Forecast Data
(GFS / ECMWF)

- 10 meteorological channels
- 10 forecast days (Day 1–10)
- Regridding & normalization
- Spatial and temporal alignment

2

Spatiotemporal Model

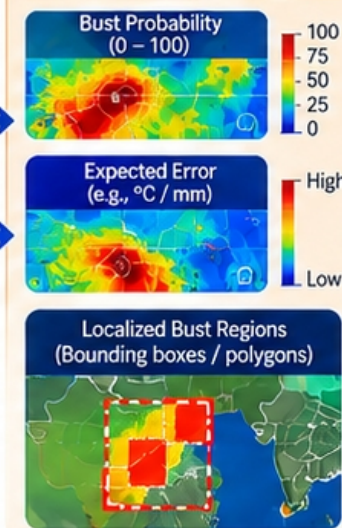
Spatiotemporal
Deep Learning Model

- U-Net + ConvLSTM
- Temporal Attention
- Spatiotemporal Feature Learning
- Captures spatial patterns + temporal evolution

3

Output Generation

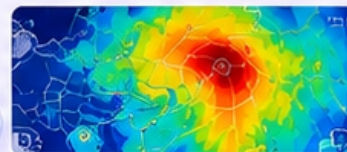
Model Outputs



4

Visualization & Alerts

Explainability & Drivers



- Key meteorological drivers (Grad-CAM + SHAP)
- Localized reasoning
- Confidence index (0–100)
- Time-wise evolution (Day 1–10)
- Alerts for high-risk regions
- Downloadable outputs (maps, GeoJSON, NetCDF)

01

Data Input & Preprocessing



- Multi-parameter meteorological data from GFS/ECMWF.
- 10 channels × 10 forecast days.
- Regridding, normalization and spatiotemporal alignment.
- Creates input tensors for the model.

02

Spatiotemporal Model



- U-Net + ConvLSTM with temporal attention.
- Learns spatial patterns and temporal evolution.
- Predicts probability and magnitude of forecast busts.
- Provides explainable features using Grad-CAM + SHAP.

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Output Generation



- Generates bust probability map (0–100).
- Outputs expected error (e.g., °C / mm).
- Identifies localized bust regions (bounding boxes / polygons).
- Provides confidence index.
- Extracts key meteorological drivers for explainability.

04

Visualization & Alerts



- Interactive map with Day 1–10 navigation.
- Layer selection (probability, error, variables).
- Localized bust region highlighting.
- Explainable AI insights.
- Threshold-based alerts for high-risk areas.
- Downloadable outputs (maps, GeoJSON, NetCDF) for further use.



Prototype / System Demonstration

AETHER-BUST provides an **interactive web-based dashboard** to visualize forecast **bust probability**, **expected error** and **meteorological explanations** for Day 1–10. The prototype demonstrates the end-to-end pipeline using synthetic/ERA5-style data (v1) with realistic weather patterns.



Key Functionalities Demonstrated

- ✓ Interactive map with bust probability and error layers
- ✓ Day 1–10 forecast navigation
- ✓ Variable selection (temperature, precipitation, winds, etc.)
- ✓ Localized bust regions (bounding boxes / polygons)
- ✓ Meteorological explanations using Grad-CAM + SHAP
- ✓ Threshold-based alerts and downloadable outputs



01

Dashboard Features



- Interactive map with Day 1–10 navigation.
- Layer selection (probability, error, confidence, variables).
- Shows localized bust regions (bounding boxes / polygons).
- Visualizes meteorological explanations (Grad-CAM + SHAP).

02

Sample Outputs (Prototype)



- Forecast bust probability map (0–100).
- Expected error map (e.g., °C / mm).
- Confidence index map.
- Localized high-risk regions.
- Top contributing meteorological drivers.
- Time-wise evolution (Day 1–10).

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User Interaction



- Select variable (t2m, tp, z500, ws850).
- Choose forecast day (1–10).
- View risk maps and error maps.
- Explore XAI explanations.
- View and filter high-risk alerts.
- Download outputs (maps, GeoJSON, NetCDF).

04

Technology Stack (Prototype)



- **Frontend:** React + Leaflet / Mapbox
- **Backend:** Python (FastAPI)
- **Model:** U-Net + ConvLSTM + Temporal Attention
- **XAI:** Grad-CAM + SHAP
- **Data:** Synthetic / ERA5-style (v1)
- **Outputs:** Maps, GeoJSON, NetCDF



Results & Performance

AETHER-BUST effectively predicts forecast busts and provides accurate **risk localization**, **expected error estimation**, and **explainable meteorological drivers**. Evaluated on synthetic / ERA5-style data (v1), the model shows significant improvement over baseline models in both spatial accuracy and error prediction.



Quantitative Results

Model	Bust Detection (AUC ↑)	Error Prediction (RMSE ↓)	Localization (IoU ↑)
AETHER-BUST (U-Net + ConvLSTM + TA)	0.91	0.28	0.62
U-Net (Spatial Only)	0.84	0.36	0.51
ConvLSTM	0.86	0.34	0.55
U-Net + LSTM	0.88	0.31	0.58
Baseline (Persistence)	0.71	0.52	0.34



Key Improvements (vs Baseline)



AUC

↑ 28%

0.71 → 0.91



RMSE

↓ 46%

0.52 → 0.28



Localization (IoU)

↑ 82%

0.34 → 0.62

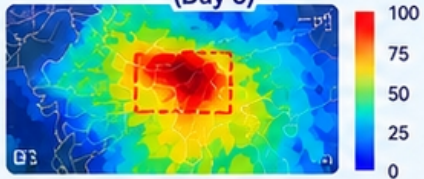
Early Detection
(High-Risk Regions)

↑ 2-3 Days

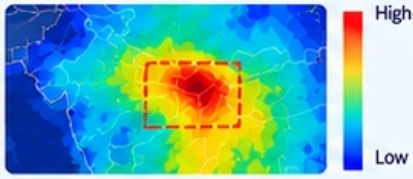
before severe deviations



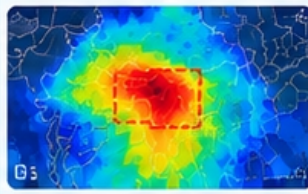
Qualitative Results (Sample Outputs)

Forecast Bust Probability
(Day 5)Localized high-risk region
(bounding box)

Expected Error (mm)

Predicted magnitude of
deviation (e.g., precipitation)

XAI Explanation (Grad-CAM)

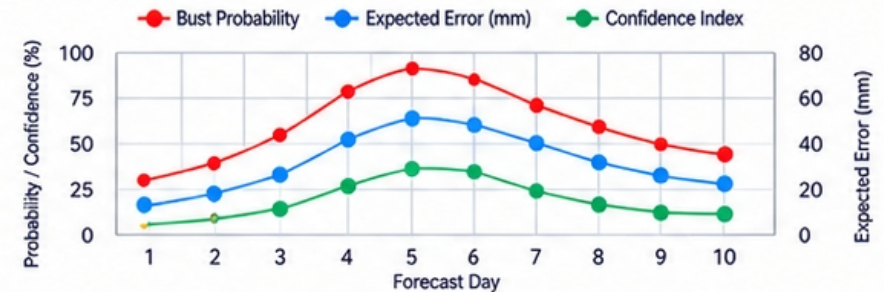
Meteorological drivers for
forecast bust (explainability)

Top Contributing Factors

Moisture (q850)	0.32
Wind Shear (ws850)	0.21
Geopotential (z500)	0.18
Temperature (t2m)	0.14
Relative Humidity	0.08



Time-wise Evolution (Selected Region)



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Operational Impact



- Early warning of forecast busts (2-3 days in advance).
- Helps disaster management agencies take proactive decisions.
- Supports agriculture, urban planning and infrastructure safety.
- Reduces socio-economic losses due to forecast failures.

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Target Users



- India Meteorological Department (IMD)
- National Disaster Management Authority (NDMA)
- State Disaster Management Authorities (SDMAs)
- Agriculture, Water Resource, and Urban Planning Departments
- Research and Academic Institutions

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Scalability & Future Scope



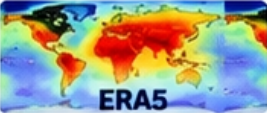
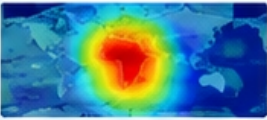
- Integration with live GFS / ECMWF data (operational deployment).
- Incorporation of additional AI models (Transformers, Graph Neural Networks).
- Higher spatial resolution and more meteorological variables.
- Real-time alert dissemination via APIs and mobile applications.
- Expansion to multiple weather hazards (cyclones, heatwaves, heavy rainfall).
- Implementing Celery message queues to scale 5D atmospheric tensor inferences for multiple concurrent operational users.

04

Societal & Environmental Benefits



- Improved disaster preparedness and resilience.
- Timely warnings for vulnerable regions.
- Optimized resource allocation during extreme weather events.
- Supports sustainable development and climate adaptation.
- Contributes to safer and climate-resilient communities.

1		ECMWF – Integrated Forecasting System (IFS) Official documentation of ECMWF's IFS and medium-range weather forecasts (up to 15 days). Provides information on model data, variables, resolution and access methods.	https://www.ecmwf.int/en/forecasts/documentation-and-support/medium-range-forecasts	Official Documentation
2		NOAA / NCEP – Global Forecast System (GFS) Official information on the Global Forecast System (GFS) model from the NCEP Environmental Modeling Center. Details on model characteristics, forecast range and data access.	https://www.emc.ncep.noaa.gov/emc/pages/numerical_forecast_systems/gfs.php	Official Documentation
3		Hersbach et al. (2020) – The ERA5 global reanalysis ERA5: the fifth generation ECMWF reanalysis for the global climate and weather. Provides high-resolution historical atmospheric data used for training and evaluation.	https://doi.org/10.1002/qj.3803	Research Paper
4		Ronneberger et al. (2015) – U-Net: Convolutional Networks for Biomedical Image Segmentation Introduces the U-Net architecture for precise spatial localization and segmentation, which forms the spatial encoder-decoder component in our model.	https://arxiv.org/abs/1505.04597	Research Paper
5		Shi et al. (2015) – Convolutional LSTM Network for Precipitation Nowcasting Proposes ConvLSTM for spatiotemporal sequence modelling, used as the temporal component in our architecture to capture time evolution in weather forecasts.	https://proceedings.neurips.cc/paper/2015/hash/07563a3fe3bbe7e3ba84431ad9d055af-Abstract.html	Research Paper
6		Selvaraju et al. (2017) – Grad-CAM: Visual Explanations from Deep Networks Introduces Grad-CAM for visualizing important regions in CNN predictions, used in our system for meteorological explainability of forecast bust regions.	https://openaccess.thecvf.com/content_iccv_2017/papers/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.pdf	Research Paper
7		Lundberg & Lee (2017) – A Unified Approach to Interpreting Model Predictions Introduces SHAP (SHapley Additive exPlanations) for feature attribution, used to identify key meteorological drivers for forecast busts.	https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html	Research Paper
 Implementation References (Prototype) Frameworks and tools used for developing the prototype (dashboard, data processing and model inference).  Python https://www.python.org/  FastAPI https://fastapi.tiangolo.com/  React https://react.dev/  MongoDB https://www.mongodb.com/  MapLibre GL JS https://maplibre.org/  GeoJSON / NetCDF https://www.earthdata.nasa.gov/				